

AI-Augmented Customer Shopping Trends Analysis

Executive Summary

This report delivers a data-driven evaluation of customer purchasing patterns, category dynamics, and loyalty structures based on transaction logs totaling **\$233,081.00** in revenue. The average purchase amount across all customer transactions stands at **\$59.76**.

The analysis was **AI-augmented**, leveraging **Generative AI** for business understanding, dataset documentation, interpretation of exploratory data analysis (EDA), and generation of business insights, alongside **SQL**, **Python**, and **Power BI** for data processing, analysis, and visualization.

Key takeaways highlight that **Clothing** is the dominant revenue generator, **Male customers** account for over two-thirds of total spend, and purchasing trends remain relatively stable across seasonal periods. Furthermore, **43.00%** of transactions involve discounts or promotional codes, while **27.00%** of customers maintain an active subscription status.

Key Findings

- **Total Revenue Generated:** **\$233,081.00** across all categories and demographic segments.
 - **Average Spend:** **\$59.76** per transaction.
 - **Dominant Category:** Clothing contributes **\$104,264.00 (44.73%)** of total revenue.
 - **Highest Average Order Value:** Footwear leads with an average purchase amount of **\$60.26**.
 - **Gender Revenue Split:** Male customers contribute **\$157,890.00 (67.74%)**, while Female customers contribute **\$75,191.00 (32.26%)**.
 - **Subscription Adoption Rate:** **27.00%** (1,053 subscribers vs. 2,847 non-subscribers).
 - **Discount / Promo Utilization Rate:** **43.00%** of transactions used a discount or promotional code (1,677 orders).
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Customer Demographics Insights

Gender Revenue Share

- **Male: \$157,890.00** (67.74%)
- **Female: \$75,191.00** (32.26%)

Insight:

Revenue generation is significantly higher among male customers, contributing more than twice the revenue generated by female customers within this dataset.

Top Geographic Markets (by Revenue)

Location	Revenue
Montana	\$5,784.00
Illinois	\$5,617.00
California	\$5,605.00
Idaho	\$5,587.00
Nevada	\$5,514.00
Alabama	\$5,261.00
New York	\$5,257.00
North Dakota	\$5,220.00
West Virginia	\$5,174.00

Nebraska	\$5,172.00
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Insight:
These locations generated the highest purchase amounts in the dataset and may represent strong-performing regional markets for further business analysis.

Product & Category Insights

Revenue Contribution

Category	Revenue	Share
Clothing	\$104,264.00	44.73%
Accessories	\$74,200.00	31.83%
Footwear	\$36,093.00	15.49%
Outerwear	\$18,524.00	7.95%

Average Order Value

Category	Average Purchase Amount
Footwear	\$60.26
Clothing	\$60.03
Accessories	\$59.84
Outerwear	\$57.17

Insight:
Although Clothing and Accessories contribute **76.56%** of total revenue, the average purchase amount remains relatively consistent across all categories.

Purchase Behavior Insights

- Average transaction value remains consistent at **\$59.76**.
 - Clothing and Accessories account for the majority of customer purchases.
 - Customer purchasing behavior appears evenly distributed across product categories, with no major variation in average order values.
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Seasonal Trends

Season	Revenue	Share
Fall	\$60,018.00	25.75%
Spring	\$58,679.00	25.17%
Winter	\$58,607.00	25.14%
Summer	\$55,777.00	23.93%

Insight:
Seasonal purchasing remains relatively stable throughout the year, with Fall generating the highest revenue and Summer showing a slightly lower contribution.

Payment Method Insights

Payment Method	Transactions	Share
PayPal	677	17.36%
Credit Card	671	17.21%

Cash	670	17.18%
Debit Card	636	16.31%
Venmo	634	16.26%
Bank Transfer	612	15.69%

Insight:

Customer payment preferences are well distributed across all available payment methods, indicating no strong dependence on a single payment channel.

Subscription & Loyalty Insights

- **Subscribers:** 1,053 (27%)
- **Non-Subscribers:** 2,847 (73%)

Insight:

A majority of customers are not enrolled in the subscription program, presenting an opportunity to explore strategies for increasing subscription adoption and customer retention.

Discount & Promo Code Insights

Promotional Usage

- Discount Applied: **1,677 (43%)**
- No Discount: **2,223 (57%)**
- Promo Code Used: **1,677 (43%)**
- No Promo Code: **2,223 (57%)**

Insight:

Discount usage aligns exactly with promotional code usage, indicating that promotional campaigns were consistently delivered through promo codes. However, since the dataset does not include profit or cost information, the financial impact of these promotions cannot be evaluated.

Customer Satisfaction Insights

Category	Average Review Rating
Footwear	3.79
Accessories	3.77
Outerwear	3.75
Clothing	3.72

Insight:
Customer satisfaction remains relatively consistent across all product categories, with Footwear receiving the highest average review rating.

Business Recommendations

1. Continue investing in **Clothing** and **Accessories**, as they contribute over **76%** of total revenue.
 2. Investigate the purchasing behavior of male customers to understand the factors contributing to their higher revenue and identify opportunities to improve engagement among female customers.
 3. Develop targeted loyalty and subscription campaigns to increase conversion among the **73%** of customers who are not currently subscribed.
 4. Evaluate promotional campaigns to determine whether discounts and promo codes improve customer engagement and purchase frequency. Since profitability data is unavailable, financial impact should be assessed using additional business metrics.
 5. Review customer feedback for the Clothing category to identify opportunities for improving customer satisfaction while maintaining its position as the highest revenue-generating category.
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Conclusion

The analysis reveals a stable retail business supported by a consistent average transaction value (**\$59.76**) and balanced seasonal purchasing patterns. Revenue is primarily driven by **male customers** and the **Clothing** and **Accessories** categories. The findings also highlight opportunities to increase subscription adoption, further analyze customer purchasing behavior, and evaluate promotional effectiveness using additional business metrics.

By combining **SQL** for data querying, **Python** for data cleaning and exploratory analysis, **Power BI** for interactive dashboard development, and **Generative AI** for business understanding, documentation, and insight generation, this project demonstrates a complete **AI-Augmented Customer Shopping Trends Analysis** workflow that supports data-driven decision-making.